

Design and implementation of a Computer Music System

Description:

Implement a complete computer music system with interaction design principles.

Main goals:

- Decide the purpose of your final system (i.e., music performance system, human-interactive effects, augmented reality synthesis, and different cool things!)
- Decide and state the tools used and the communication protocols for sharing information
- Implement your computer music system

Tools:

Each project needs to satisfy at least the block diagram presented below. You can expand your solution with other blocks and connections. However, the following three blocks are required:

- interaction system unit

You can use a device of your preference to capture the user inputs (i.e., Arduino, Bela, Mobile phone, etc...).

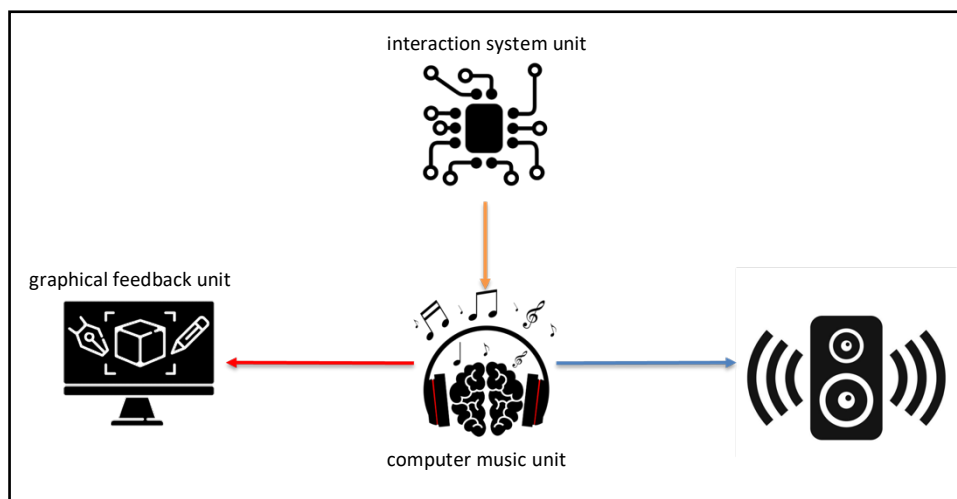
- computer music unit

You must use “SuperCollider” and “JUCE”.

The normal configuration is that the synthesis is performed in one of the two environments, whereas effects are implemented in the other one. Anyway, if you wish to propose a different scheme, please go ahead. This will be considered by the instructors.

- graphical feedback unit

Implement a GUI with “Processing” to provide some graphical feedback to the user.



NB:

Do not focus on complicated things, but on original ones!

Be creative, feel free to exploit interesting solutions!

 POLITECNICO MILANO 1863	Academic Year 2025-2026 052483 Computer Music: Languages and Systems	Homework assignment
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Output:

- A brief presentation and demonstration of your work that will be given to the class. (duration of the presentation will be communicated later)
- A public repository in the general GitHub channel (URL to be shared in a few days). Add in your repo a general README file with a brief description of the project. Identify the repo as specified in the guidelines to be distributed soon.

Procedure and deadlines

- Formalize your proposal by April 5, 2025 through the online form available here: [CMSL A.Y. 2025-2026 - HW proposal – Fill out form](#) I don't expect by that time that the interaction design part is completely defined, but that you give us a general idea of how the user will interact with the computer music system.
- Reserve the devices and sensors (1 board, 1 digital sensor, 1 analog sensor) by April 27, 2025 through the following online form: [CMLS - Material booking \(Copy\) – Fill out form](#).
- Upload on the GitHub channel the report and the code by May 26, 2025, 11:59 pm CET.
- Presentation on May 27 (MI students) and June 03 (CR students).